

SLR Parsing Table

09 April, 2026

Construction of SLR parsing table

Input: An augmented grammar G' .

Output: The SLR parsing table functions ACTION and GOTO for G' .

Method:

1. Construct $C = \{I_0, I_1, I_2, \dots, I_n\}$, the collection of sets of LR(0) items for G' .
2. State i is constructed from I_i . The parsing actions for state i are determined as follows:
 - a. If $[A \rightarrow \alpha \cdot a\beta]$ is in I_i and $\text{GOTO}[I_i, a] = I_j$, then set $\text{ACTION}[i, a]$ to "Shift j ". Here, a must be a terminal.
 - b. If $[A \rightarrow \alpha \cdot]$ is in I_i , then set $\text{ACTION}[i, a]$ to "Reduce $A \rightarrow \alpha$ " for all in a in $\text{FOLLOW}(A)$.
 - c. If $[S' \rightarrow S \cdot]$ is in I_i , then set $\text{ACTION}[i, \$]$ to ACCEPT.

If any conflicting actions are generated by the above rules, we say that the grammar is SLR(1), in which case the algorithm fails to produce a parser.

3. The GOTO transitions for state i is constructed for all non-terminals A using the rule:
If $\text{GOTO}[I_i, A] = I_j$, then $\text{GOTO}[i, A] = j$.
4. All entries not defined by rules 2 and 3 are made ERROR.
5. The initial state of the parser is the one constructed from the set of items containing $[S' \rightarrow \cdot S]$.